PRD - Similar Generated Questions v1

Overview

Requesting Client:

Zenkyoken, Renseikai

Target Market/User Segment:

Students preparing for exams, across all school segments from Junior High upwards.

Problem Statement

When students are preparing for exams, they want to practice questions in the areas where they are not confident. However, existing problem banks are limited in size and scope, and often do not focus deeply on the exact concepts a student is struggling with.

As a result, students frequently hit a learning wall: they run out of relevant practice material before they feel confident.

The Similar Question Generation feature removes this barrier by allowing students to generate as many targeted, similar questions as they need, enabling focused practice until they feel fully confident.

This unlocks continuous, personalized practice at the exact point of need.

Competitor & Market Intelligence

Atama+ and Knock (need to double-check Knock), our closest competitors, provide a "similar questions" feature. However, for Atama+, we believe they have a bank of similar questions that they can pull out for the student to practice, which has limitations around depth and coverage.

Success Metrics

- Number of generated similar questions per student
  - This directly indicates the amount of practice students are doing through AI Tutor
  - extra info:
    - correct / incorrect / unanswered
    - timestamp (so we can see if practice is leading to correct or not)

Hero Loop

The core user journey supported by this feature is:

Snap → Learn → Practice → Move on

1. Student takes a photo of a problem (Snap)

2. Student learns from the hints/solution (Learn)
3. Student practices with generated similar questions (Practice)
4. Student proceeds to the next concept or problem (Move on)

Where This PRD Fits in the Loop

This feature directly powers the **Practice** step of the learning loop.

The Wow Moment

Students can practice exactly what they're struggling with, as many times as they need - until they confident.

Additional Note

Post ToC Integration, the core loop becomes:

Review Past Mistake → Practice → Move on

Functional Requirements

At this stage, similar question generation is always initiated at the **question level**. After ToC integration, we will offer similar question generation at the topic/chapter level (covering multiple questions at once).

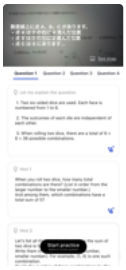
Entry Point – Question Level

User Story:
As a learner, I want to generate similar questions from the specific question I am viewing so that I can practice more of the same type.

Acceptance Criteria:

- From any question view, the user can start similar question generation
- Generation is always tied to the currently viewed question
- The user taps **Practice** on a snap to begin the flow

UI Reference:



Button title to use: Practice (no sub-print) / 類題作成

[Figma node link]

Practice Mode Entry Independent of Hint/Solution

Requirement: Practice should be available immediately on Snap page

User Story
As a learner, I want to start Similar Question Practice immediately after opening a snap, without waiting for hint/solution generation.

Acceptance Criteria

- Practice entry is available as soon as the Snap page is displayed
- Practice entry does not depend on hint/solution generation or rendering completion
- Student can generate similar questions even while hints/solution are still loading

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Format Constraint for Generated Questions

Requirement: Question Format

- All similar generated questions must be **multiple choice format only**

Acceptance Criteria:

- Every generated question presented to the user is multiple choice
- No free-response or open-ended questions are generated at this time
- 4 options always

UI Reference:

[Figma node link]

Generation Prompt Structure

Similar question generation will use the following approach:

- Rely on **additional context for curriculum knowledge**
- No additional curriculum databases or external knowledge sources are required

Prompt Reference:

- Final prompt and context flow is currently being worked on by the PS team
- It will be finalized over the next few weeks

Prompt Link:
`[TBD]`

~~Sub-question Context Handling~~

~~**Requirement:** Context-aware generation for sub-questions without hierarchy metadata~~

~~**User Story**~~
~~As a learner, when I generate a similar question for a sub-question, I want the system to consider the parent and related prior question context so that the generated question is accurate.~~

~~**Acceptance Criteria**~~

- ~~When generation is triggered for a target question, the request includes:~~
  - ~~the target sub question~~
  - ~~the **previous question & answer** within the same snapped question set, if it exists~~
    - ~~if answer is not generated, leave blank~~
- ~~The generated output must still conform to MCQ format requirements~~

~~**Note (this is V1 behavior)**~~
~~Because hierarchy metadata does not yet exist, V1 uses "the previous question" as context.~~

Loading Experience During Generation

Requirement: Loading Page

Once the user initiates similar question generation, the next step in the journey is waiting for the question to be created.

Acceptance Criteria:

- After the user initiates similar question generation, a loading page is displayed
- The loading page uses messaging similar to the existing 'analyzing question' loader
- The displayed text must state:
  “Creating a similar question”
- Once generation finishes, the first similar question is displayed automatically

UI Reference:

[redacted]

Background Continuation During Loading

User Story

As a learner, if I minimize/close/leave while a similar question is loading, I want generation to continue so I can resume without losing progress.

Acceptance Criteria

- If user minimizes app, closes app, or navigates away during loading, generation continues in background
- Existing background mechanism used for hint generation is reused
- On return, the student sees the latest resolved state:
  - generated question if complete
  - loading state if still in progress
  - recoverable error state if failed

Core Practice Flow

User Story:

As a learner, I want to answer generated similar questions so that I can practice the concept.

Acceptance Criteria:

- The system presents a generated similar question
- User can select an answer (multiple choice)
- Immediate feedback is shown after the user selects an answer

Feedback Behavior:

- If the user selects the correct answer:
  - The selected option is highlighted in **green**
  - A success indicator is shown
  - An explanation is displayed describing why the answer is correct
- If the user selects an incorrect answer:
  - The selected option is highlighted in **red** to indicate it is wrong
  - The correct answer option is highlighted in **green**
  - An explanation is displayed describing why the correct answer is correct
- A button labeled **"One More"** is available after feedback
  - Tapping the button generates the next similar question based on the current question
  - User can tap One More **even when the current generated question is *unanswered***

UI Reference:

Button title to use: One More / 追加作成

[Figma node link]

Answer Locking (Immutability)

User Story

> As a learner, once I submit an answer to a generated question, I should not be able to change that answer accidentally.

Acceptance Criteria

- After the first answer selection, that answer becomes locked
- Tapping other options after submission does not change the result
- Feedback and explanation remain unchanged after locking
- Navigating away and returning to the same question preserves the locked state

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Navigation Between Generated Questions

User Story:

As a learner, I want to navigate between generated questions so that I can review previous ones.

Acceptance Criteria:

- When more than one similar question has been generated:
  - A **Previous** button is shown
  - User can move back to earlier generated questions
  - The **One More** button will only appear on the last generated question
    - Otherwise a **Next** button is shown in its place for navigation

UI Reference:

Button title to use: **Previous/前**、**Next/次**OR **One More/追加作成**

[Figma node link]

Dynamic Question Index Display

User Story

> As a learner, I want to see which generated question I am currently on so that I can understand my position within the practice session.

Acceptance Criteria

- A question index indicator is visible during practice mode (e.g., “1/3”)
- The index increments correctly when a new question is generated
- The index updates properly when navigating using Previous or Next
- The index remains accurate after answering a question
- The index resets appropriately when a new practice session is started

UI Reference

[Figma node link]

Exit Practice Mode

Requirement: Back Navigation

This is the final step in the journey, allowing the learner to leave practice mode and return to normal study flow.

Acceptance Criteria:

- A back button is always available during practice mode
- Selecting the back button exits the similar question flow
- The user is returned to the previous screen from which practice was launched

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Persistent Practice History Per Question

User Story

As a learner, when I return to a question I practiced before, I want to see prior generated questions and continue where I left off.

Acceptance Criteria

- Practice history is stored **per source question**
- Re-entering practice for the same source question shows prior generated questions
- The **latest generated question** is displayed by default on entry
- If the latest question was unanswered, it remains unanswered and can be answered on return
- If generation was in progress when user left, user resumes to in-progress/completed result state on return
- Session behavior changes from “forget-and-reset” to “continuous history”

V1 feasibility note

- Intended for V1 **if implementation cost remains acceptable**; if not, must be explicitly descoped.

Retry Behavior on Generation Error

User Story

> As a learner, if similar question generation fails, I want the system to retry once automatically so that I do not have to restart the process myself.

Acceptance Criteria

- If a recoverable error occurs during generation, the system automatically retries once
- The loading screen remains visible during the retry
- If the retry succeeds, the generated question is displayed normally
- If the retry fails, the user is shown a clear, recoverable error state
- The system performs at most one automatic retry per generation request

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Manual Retry Button on Generation Error (After Automatic Retry)

User Story

> As a learner, if similar question generation fails even after the system has already retried automatically, I want a clear Retry button so that I can attempt to generate another question without leaving practice mode.

Acceptance Criteria

- If generation fails, the system first performs **one automatic retry**
- If the automatic retry succeeds, the generated question is displayed normally and no error state is shown
- If generation fails again after that automatic retry, the user is shown an error state
- The error state includes a visible **Retry** button
- Tapping the Retry button restarts the generation process from scratch
- When Retry is tapped, the loading screen is shown again
- If this manual retry succeeds, the new generated question is displayed normally
- If it fails again, the same error state remains and the user can attempt Retry again
- The user can exit practice mode using the Back button from the error state at any time

No UI Reference at this time (Developer best effort)

Functional Requirement: Teacher Tracking (By Student, By Question)

Requirement: Track practice analytics for teachers at student-question level with time trend.

User Story

As a teacher, I want to track each student's similar-question practice by question over time so I can diagnose confidence and support intervention.

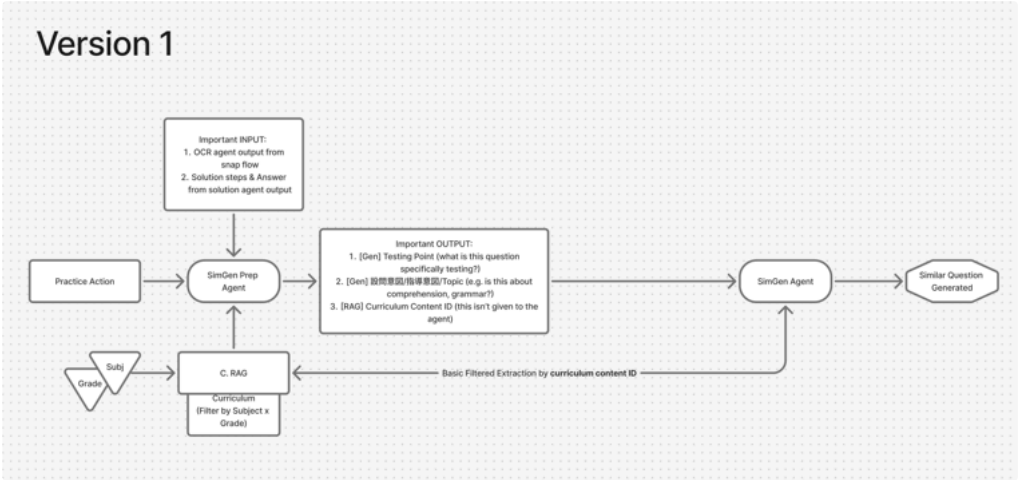
Acceptance Criteria

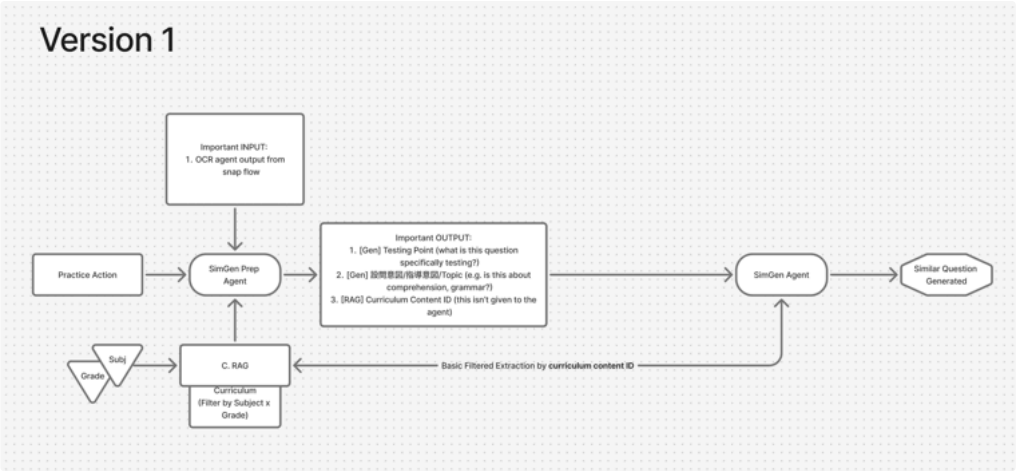
- Track metrics **by student + source question**:
  - question explanation/summary
  - number of similar questions generated
  - number of correct attempts
  - number of incorrect attempts
  - number of unanswered attempts
- All tracked events are timestamped to support trend analysis over time (to answer the question: *after practicing, is the student mastering this question or still stuck?*)

Non-Functional Requirements

- Aim for <10 seconds for generations

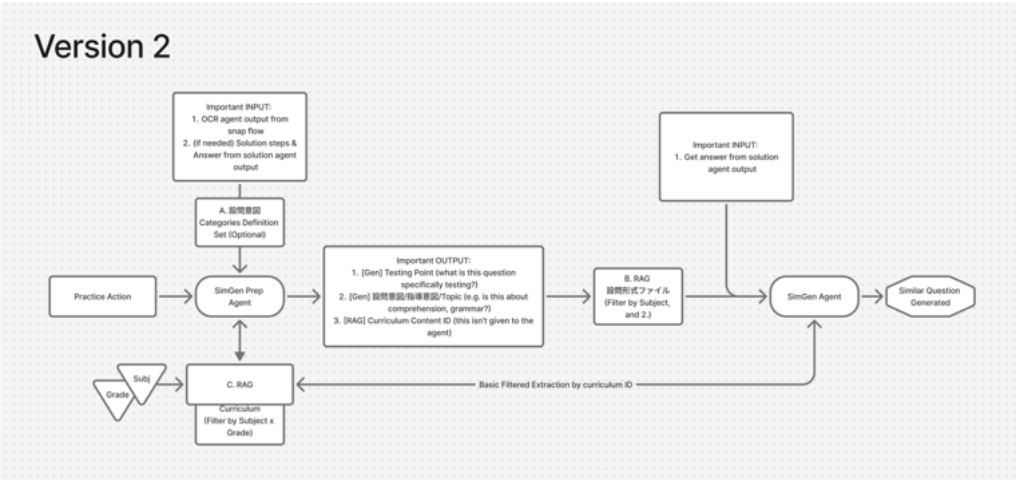
Context Flows





指導意図→Teaching Intent

Later:



Test Cases

Test Cases

Future Considerations

The following items are intentionally deferred and prioritized for potential future phases.

Related-Question Metadata for Sub-questions - Confirmed

- Introduce metadata to identify related parent/sub-question groupings
- Use relatedness metadata to avoid sending unrelated context

- Consider generating related sub-questions as a grouped unit instead of one-by-one context packing

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Table of Contents (ToC) Integration – Confirmed

In the future:

- Integrate similar question generation directly with the Table of Contents experience
- Allow students to initiate practice from topic or chapter level
- Support workflows such as:
  - Reviewing past mistakes from ToC
  - Jumping directly into similar question practice from a specific topic area

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Practice Statistics and Progress Tracking – Confirmed

In the future:

- Display statistics around similar question practice
- Track number of attempts and number of correct answers
- Provide aggregated progress indicators to help students understand confidence and mastery

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User-Controlled Quantity of Similar Questions – Not Confirmed

In the future:

- Users should be able to **choose how many similar questions to generate**
- Provide UI controls to set this preference before generation

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Conversational Support for Generated Questions – Not Confirmed

In the future:

- Users should be able to **chat regarding a similar generated question**
- Ask follow-up questions or request additional explanations directly within practice mode

Spec Sharing Session 2/10: Memo

- For a question with multiple sub questions, when we generate similar questions for a sub question (since we currently don't have hierarchy in questions), we need to consider the context of the parent question and related sub questions so that we generate accurately. (Trieu)
  - first, send the previous questions as context (James)

        - only need to send the previous question & answer as context (Hida, post meeting chat)
    - in future: we need to generate new meta for questions to say they are related, so that we don't feed unrelated questions to generate a sub question. In fact, we should probably consider generating related sub questions together. (John)
- *if needed*, feed in up to 5 previous 'similar' questions to ensure the next similar question is 'different enough' (Cuong)
- make sure to define what we need to track for teachers (Cuong):
    - by student, by question (explanation):
        - number of similar questions generated
        - number of correct attempts
        - trended over time
- if user leaves the app while loading, it should continue in the background (James). Reuse mechanism for hint (Trieu)
- Change it to keep the history of practice questions per question, if user clicks practice on a question they practiced before, the previous ones will be there, and so will the One More button so they can continue to practice (James). Show last one by default so the user doesn't have to navigate before accessing the One More button (Trieu).
    - In this way, we change the requirement from forget-and-reset every practice session, to a continuous experience - as long as dev cost doesn't get too high for v1 (John).
- Allow One More similar question generation without answering the current one (James)
    - Consider the loading button so student can generate the next one while working on the current one. This is so that if the AI generated a bad answer, they can skip to answer (John)
- Show the practice button before hint/solution is generated (John)